LoRA vs. Full Fine-Tuning

A Theoretical Perspective on Generalization

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LoRA · Generalization · Theory

TL;DR We derive **excess-risk bounds** for LoRA and Full FT in linear regression. In both the over- and under-determined regimes, LoRA reduces variance from $\Theta(d_x d_y/n)$ to $\mathcal{O}(r\max\{d_x,d_y\}/n)$. When the task shift $\mathbf{\Delta}^\star$ is effectively **low-rank**, LoRA **strictly outperforms** FFT; the gap is largest in *high-noise / few-sample* regimes. Qwen2.5 fine-tuning experiments confirm the linear theory.

1 Problem setup

Linear regression with n i.i.d. samples $\mathcal{D} = \{(\mathbf{x}_i, \mathbf{y}_i)\}_{i=1}^n$:

$$\mathbf{y}_i = \mathbf{A}^\star \mathbf{x}_i + \boldsymbol{\varepsilon}_i, \quad \boldsymbol{\varepsilon}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{\Sigma}_{\varepsilon\varepsilon}), \quad \mathbf{\Delta}^\star := \mathbf{A}^\star - \mathbf{A}_0.$$

Stacking samples as columns gives the data matrices:

$$\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_n] \in \mathbb{R}^{d_x \times n}, \qquad \mathbf{Y} = [\mathbf{y}_1, \dots, \mathbf{y}_n] \in \mathbb{R}^{d_y \times n}.$$

Both methods solve $\min_{\mathbf{A}} \|\mathbf{AX} - \mathbf{Y}\|_{\mathrm{F}}^2$. They differ only in the constraint on $\mathbf{A}$, and we always pick the *min-norm* solution.

FFT (min-norm)

$$\widehat{\mathbf{A}}_{\mathrm{FFT}} = \mathbf{A}_0 + \widehat{\mathbf{\Delta}}_{\mathrm{FFT}}, \qquad \widehat{\mathbf{\Delta}}_{\mathrm{FFT}} := (\mathbf{Y} - \mathbf{A}_0\mathbf{X})\,\mathbf{X}^\dagger.$$

LoRA (rank-r, Sondermann)

$$\widehat{\mathbf{A}}_{\mathrm{LoRA}} = \mathbf{A}_0 + \mathrm{trunc}_r\{\widehat{\mathbf{\Delta}}_{\mathrm{FFT}}\,\widehat{\mathbf{\Sigma}}_{\mathbf{xx}}^{1/2}\}\,(\widehat{\mathbf{\Sigma}}_{\mathbf{xx}}^{1/2})^\dagger.$$

LoRA's update is the FFT update *projected onto its top-r singular components* in the data geometry, then mapped back. Both estimators have closed form; we study their *expected excess risk*.

Lemma 5 (Excess-risk identity)

$$\mathcal{R}(\widehat{\mathbf{A}}) - \mathcal{R}(\mathbf{A}^\star) = \|(\widehat{\mathbf{A}} - \mathbf{A}^\star)\,\mathbf{\Sigma}_{\mathbf{xx}}^{1/2}\|_{\mathrm{F}}^2.$$

2 Overdetermined regime $n \geq d_x$

Theorem 1 (FFT excess risk)

$$\begin{aligned}\mathbb{E}[\mathcal{R}(\widehat{\mathbf{A}}_{\mathrm{FFT}})] - \mathcal{R}(\mathbf{A}^\star) &= \tfrac{1}{n}\mathrm{Tr}(\mathbf{\Sigma}_{\varepsilon\varepsilon})\,\mathbb{E}[\mathrm{Tr}(\mathbf{\Sigma}_{\mathbf{xx}}\,\widehat{\mathbf{\Sigma}}_{\mathbf{xx}}^{-1})] \\ &\geq \bar{\sigma}_\varepsilon^2\,\frac{d_x d_y}{n}, \quad \bar{\sigma}_\varepsilon^2 := \tfrac{1}{d_y}\,\mathbb{E}[\|\boldsymbol{\varepsilon}\|^2].\end{aligned}$$

When $n \geq d_x$, FFT coincides with OLS: it does not depend on $\mathbf{A}_0$ and is *unbiased*. The bound scales as $\Omega(d_x d_y/n)$ and blows up at the interpolation threshold $n \to d_x$.

Proposition 3 (Gaussian case, FFT)

If $\mathbf{x}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{\Sigma}_{\mathbf{xx}})$ i.i.d. and $\mathbf{\Sigma}_{\varepsilon\varepsilon} \succeq \mathbf{0}$:

$$\mathbb{E}[\mathcal{R}(\widehat{\mathbf{A}}_{\mathrm{FFT}})] - \mathcal{R}(\mathbf{A}^\star) = \mathrm{Tr}(\mathbf{\Sigma}_{\varepsilon\varepsilon})\,\frac{d_x}{n - d_x - 1}.$$

Theorem 2 (LoRA excess risk)

$$\begin{aligned}\mathbb{E}[\mathcal{R}(\widehat{\mathbf{A}}_{\mathrm{LoRA}})] - \mathcal{R}(\mathbf{A}^\star) \lesssim\ & \frac{r\max\{d_x,d_y\}}{n}\,\lambda_{\max}(\mathbf{\Sigma}_{\varepsilon\varepsilon})\,\mathbb{E}[\lambda_{\max}(\mathbf{\Sigma}_{\mathbf{xx}}\widehat{\mathbf{\Sigma}}_{\mathbf{xx}}^{-1})] \\ & + \lambda_{\max}(\mathbf{\Sigma}_{\mathbf{xx}})\left(r\sigma_{r+1}^2(\mathbf{\Delta}^\star) + \sum_{i>r}^{\min\{d_x,d_y\}} \sigma_i^2(\mathbf{\Delta}^\star)\right) \\ & + r\sigma_{r+1}^2(\mathbf{\Delta}^\star)\,\mathbb{E}[\lambda_{\max}(\widehat{\mathbf{\Sigma}}_{\mathbf{xx}})\lambda_{\max}(\mathbf{\Sigma}_{\mathbf{xx}}\widehat{\mathbf{\Sigma}}_{\mathbf{xx}}^{-1})].\end{aligned}$$

The first term is LoRA's *variance*. The remaining two terms are the *bias* incurred by the rank-r truncation: this error is *inevitable*, since any rank-r adaptation must misfit the spectral tail of $\mathbf{\Delta}^\star$ (terms involving $\sigma_{r+1}(\mathbf{\Delta}^\star)$ and $\sum_{i>r}\sigma_i^2(\mathbf{\Delta}^\star)$). The bias is small exactly when $\mathbf{\Delta}^\star$ has rapidly decaying singular values.

Proposition 4 (asymptotic LoRA variance)

Under the Gaussian setting of Prop. 3, the variance term in Thm. 2 is asymptotically equivalent to

$$r\max\{d_x,d_y\}\big(\sqrt{n} - \sqrt{d_x}\big)^{-2}\lambda_{\max}(\mathbf{\Sigma}_{\varepsilon\varepsilon}).$$

Far from the interpolation threshold ($n \gg d_x$), with bounded $\lambda_{\max}(\mathbf{\Sigma}_{\varepsilon\varepsilon})$ but non-vanishing $\bar{\sigma}_\varepsilon^2$ (e.g. isotropic noise), this is $\mathcal{O}(r\max\{d_x,d_y\}/n)$, versus FFT's $\Omega(d_x d_y/n)$. Since r is typically much smaller than $\min\{d_x,d_y\}$, LoRA achieves a *substantial reduction in variance*.

Corollary 1 (low-rank shift)

If $\mathrm{rank}(\mathbf{\Delta}^\star) \leq r$:

$$\mathbb{E}[\mathcal{R}(\widehat{\mathbf{A}}_{\mathrm{LoRA}})] - \mathcal{R}(\mathbf{A}^\star) \lesssim \frac{r\max\{d_x,d_y\}}{n}\,\lambda_{\max}(\mathbf{\Sigma}_{\varepsilon\varepsilon})\,\mathbb{E}[\lambda_{\max}(\mathbf{\Sigma}_{\mathbf{xx}}\widehat{\mathbf{\Sigma}}_{\mathbf{xx}}^{-1})].$$

Bias-variance tradeoff. As r grows, the truncation bias decreases but the variance grows linearly in r. When the spectrum of $\mathbf{\Delta}^\star$ decays fast, even a small r pays a tiny bias and reaps the full variance reduction. When the spectrum is flat, no $r < \min\{d_x,d_y\}$ can compete with FFT. In modern LLMs the spectrum is heavily concentrated in a few leading directions, making the low-rank assumption mild.

FFT vs. LoRA. $\Theta(d_x d_y/n)$ *vs.* $\mathcal{O}(r\max\{d_x,d_y\}/n)$*: when* $r \ll \min\{d_x,d_y\}$*, LoRA achieves a substantial variance reduction.*

3 Underdetermined regime $n < d_x$

Theorem 3 (FFT excess risk)

$$\mathbb{E}[\mathcal{R}(\widehat{\mathbf{A}}_{\mathrm{FFT}})] - \mathcal{R}(\mathbf{A}^\star) = \underbrace{\mathbb{E}[\|\mathbf{\Delta}^\star\mathbf{P}_{\mathbf{X}}^{\perp}\mathbf{\Sigma}_{\mathbf{xx}}^{1/2}\|_{\mathrm{F}}^2]}_{\text{FFT Bias}} + \underbrace{\tfrac{1}{n}\mathrm{Tr}(\mathbf{\Sigma}_{\varepsilon\varepsilon})\,\mathbb{E}[\mathrm{Tr}(\mathbf{\Sigma}_{\mathbf{xx}}\widehat{\mathbf{\Sigma}}_{\mathbf{xx}}^{\dagger})]}_{\text{FFT Variance}}.$$

Prop. 5 (Gaussian case, FFT)

$$\mathrm{Bias} = \|\mathbf{\Delta}^\star\|_{\mathrm{F}}^2\,\frac{d_x - n}{d_x}, \quad \mathrm{Variance} = \mathrm{Tr}(\mathbf{\Sigma}_{\varepsilon\varepsilon})\,\frac{n}{d_x - n - 1}.$$

FFT is now *biased*: the data only constrain $\mathbf{A}$ in the row space of $\mathbf{X}$, and the rest of $\mathbf{\Delta}^\star$ leaks into the risk. Bias decreases linearly with n, while variance grows with n; they balance near $n \to d_x$.

Theorem 4 (LoRA excess risk)

$$\begin{aligned}\mathbb{E}[\mathcal{R}(\widehat{\mathbf{A}}_{\mathrm{LoRA}})] - \mathcal{R}(\mathbf{A}^\star) \lesssim\ & \mathbb{E}[\|\mathbf{\Delta}^\star\mathbf{P}_{\mathbf{X}}^{\perp}\mathbf{\Sigma}_{\mathbf{xx}}^{1/2}\|_{\mathrm{F}}^2] \\ & + \frac{r\max\{n,d_y\}}{n}\,\lambda_{\max}(\mathbf{\Sigma}_{\varepsilon\varepsilon})\,\mathbb{E}[\lambda_{\max}(\mathbf{\Sigma}_{\mathbf{xx}}\widehat{\mathbf{\Sigma}}_{\mathbf{xx}}^{\dagger})] \\ & + \lambda_{\max}(\mathbf{\Sigma}_{\mathbf{xx}})\left(r\sigma_{r+1}^2(\mathbf{\Delta}^\star) + \sum_{i>r}^{\min\{n,d_y\}} \sigma_i^2(\mathbf{\Delta}^\star)\right) \\ & + r\sigma_{r+1}^2(\mathbf{\Delta}^\star)\,\mathbb{E}[\lambda_{\max}(\widehat{\mathbf{\Sigma}}_{\mathbf{xx}})\lambda_{\max}(\mathbf{\Sigma}_{\mathbf{xx}}\widehat{\mathbf{\Sigma}}_{\mathbf{xx}}^{\dagger})].\end{aligned}$$

The first term is the *same* bias as FFT (Thm. 3). The last two are truncation error from the rank-r constraint, which vanish if $\mathrm{rank}(\mathbf{\Delta}^\star) \leq r$.

Proposition 6 (asymptotic LoRA variance)

Assume $d_x > n + 1$, $\mathbf{x}_i \sim \mathcal{N}(\mathbf{0}, \sigma_x^2\mathbf{I})$ i.i.d., and $\mathbf{\Sigma}_{\varepsilon\varepsilon} \succeq \mathbf{0}$. The variance term in Thm. 4 is asymptotically equivalent to

$$r\max\{n,d_y\}\big(\sqrt{d_x} - \sqrt{n}\big)^{-2}\lambda_{\max}(\mathbf{\Sigma}_{\varepsilon\varepsilon}).$$

Corollary 2 (low-rank shift)

If $\mathrm{rank}(\mathbf{\Delta}^\star) \leq r < n$:

$$\mathbb{E}[\mathcal{R}(\widehat{\mathbf{A}}_{\mathrm{LoRA}})] - \mathcal{R}(\mathbf{A}^\star) \lesssim \mathbb{E}[\|\mathbf{\Delta}^\star\mathbf{P}_{\mathbf{X}}^{\perp}\mathbf{\Sigma}_{\mathbf{xx}}^{1/2}\|_{\mathrm{F}}^2] + \frac{r\max\{n,d_y\}}{n}\,\lambda_{\max}(\mathbf{\Sigma}_{\varepsilon\varepsilon})\,\mathbb{E}[\lambda_{\max}(\mathbf{\Sigma}_{\mathbf{xx}}\widehat{\mathbf{\Sigma}}_{\mathbf{xx}}^{\dagger})].$$

Bias parity, variance gain. *LoRA inherits FFT's bias exactly, but enjoys the $r/\min\{d_x,d_y\}$ variance reduction. When $\mathbf{\Delta}^\star$'s spectrum decays fast, LoRA Pareto-dominates FFT in both regimes.*

4 Linear regression experiment

Isotropic Gaussian features; $\mathbf{\Delta}^\star = \frac{1}{\sqrt{r^\star}}\mathbf{UV}$ with Gaussian $\mathbf{U}, \mathbf{V}$. 100 seeds; shaded ±1 s.d.

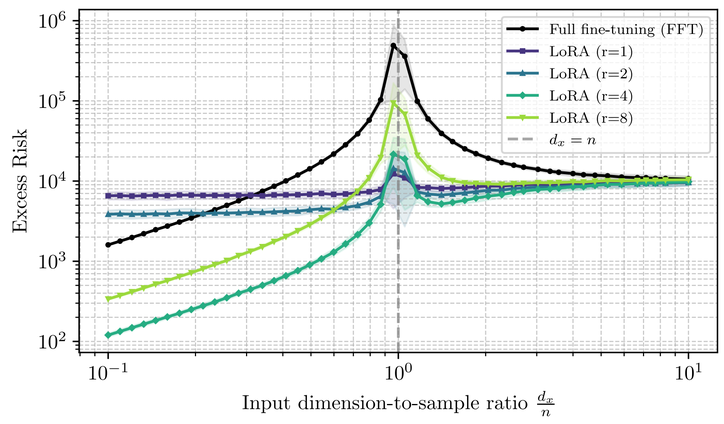


Fig. 1. *Top: excess risk vs. d_x/n with $r^\star = 4$. Bottom: excess risk vs. noise $\sigma \in [1, 100]$, $d_x = d_y = 100$, $n = 50$, $r^\star = 10$.*

- $r \geq r^\star$: LoRA beats FFT at every d_x/n; the optimum is $r = r^\star$.
- $r < r^\star$: rank-bias dominates and LoRA can hurt; risk is nearly flat in d_x/n.
- **High noise**: FFT explodes with σ; low-rank LoRA stays bounded.
- **Interpolation peak** at $d_x/n = 1$ matches the $\sqrt{n} - \sqrt{d_x}$ scaling of Props. 4 and 6.

5 LLM experiment: label noise

Qwen2.5-0.5B fine-tuned with various ranks on *BoolQ* & *CommonsenseQA*; 3 seeds. The noise level controls the variance term in our excess-risk bounds: larger $\mathrm{Tr}(\mathbf{\Sigma}_{\varepsilon\varepsilon})$ means the variance gap between FFT and LoRA dominates.

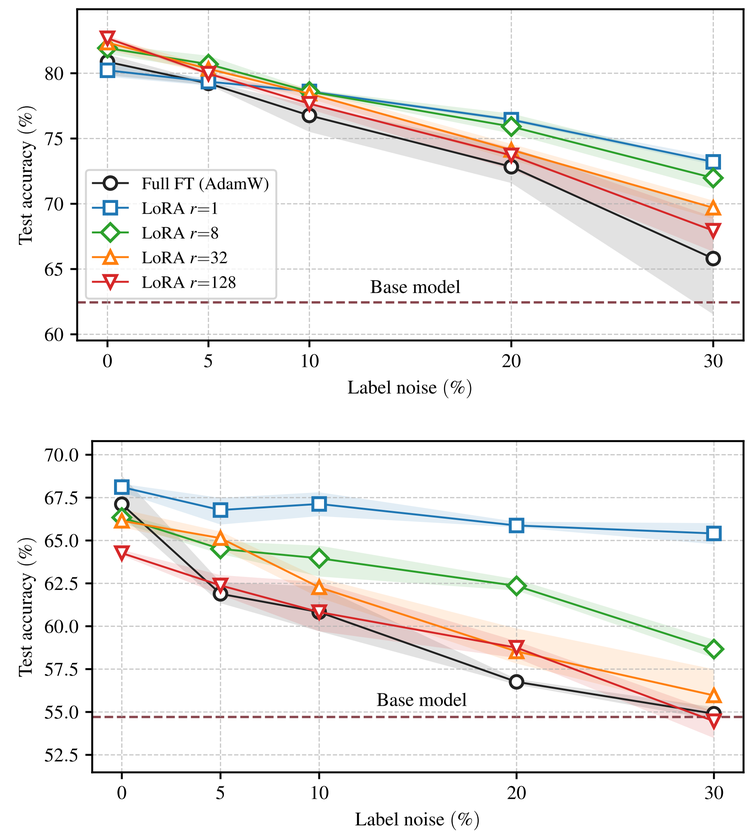


Fig. 2. *Test accuracy vs. label noise on BoolQ (top) and CommonsenseQA (bottom). Strong-noise regime: LoRA outperforms Full FT (Cor. 1).*

- **Weak noise.** No consistent rank ordering: a moderate r is optimal, the classical bias/variance tradeoff.
- **Strong noise.** Test accuracy decreases *monotonically* with r. Restricting the number of trainable directions improves generalization.
- **Why?** Variance dominates the excess risk and LoRA scales it down by $r/\min\{d_x,d_y\}$ exactly as Cor. 1 predicts.

Two regimes (theory). When $\mathrm{Tr}(\mathbf{\Sigma}_{\varepsilon\varepsilon}) \gg \|\mathbf{\Delta}^\star\|_{\mathrm{F}}^2$ the variance term dominates the bias and adding samples can hurt. When $\mathrm{Tr}(\mathbf{\Sigma}_{\varepsilon\varepsilon}) \ll \|\mathbf{\Delta}^\star\|_{\mathrm{F}}^2$ the bias governs the risk. The plots above span both regimes.

6 LLM experiment: sample size & spectrum

Qwen2.5-1.5B with various LoRA ranks across training-set sizes. Lower panels: normalized cumulative spectral energy of the estimated $\mathbf{\Delta}^\star$.

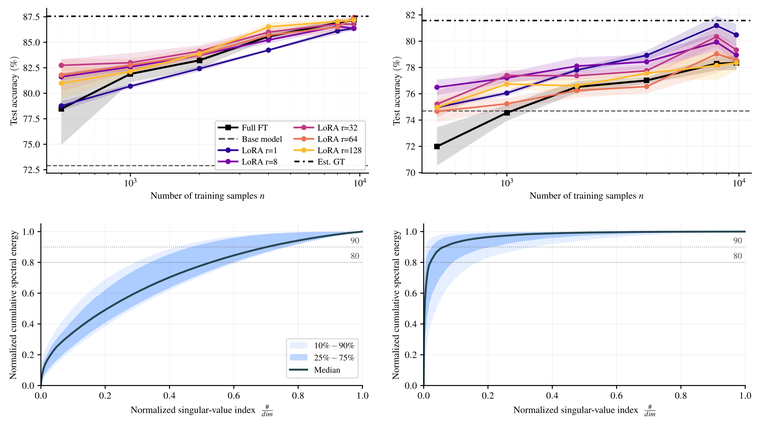


Fig. 3. *Top: accuracy vs. n. Bottom: cumulative spectral energy of $\mathbf{\Delta}^\star$. BoolQ (left): heavier tail. CommonsenseQA (right): rapid decay ⇒ $\mathbf{\Delta}^\star$ effectively low-rank.*

- **CommonsenseQA.** Rapid spectral decay, so $\mathbf{\Delta}^\star$ is effectively low-rank, and LoRA *significantly* outperforms FFT.
- **BoolQ.** Heavy spectral tail, so $\mathbf{\Delta}^\star$ is poorly approximated by rank r, and LoRA matches FFT but does not beat it.
- **Spectral decay = practical knob.** The cumulative-energy curve of an estimated $\mathbf{\Delta}^\star$ is a cheap diagnostic for when LoRA will help.
- **Confirmation across model sizes.** The qualitative trends in Figs. 2 and 3 reproduce on Qwen2.5-0.5B / 1.5B / 3B (App. VI), supporting the linear theory.

Takeaways

- First **excess-risk analysis of LoRA** covering both over- and under-determined regimes.
- FFT $\Theta(d_x d_y/n)$ vs. LoRA $\mathcal{O}(r\max\{d_x,d_y\}/n)$: variance reduced by $r/\min\{d_x,d_y\}$.
- LoRA **strictly outperforms FFT** when $\mathbf{\Delta}^\star$ is effectively low-rank; gap grows in high-noise / few-sample regimes.
- Qwen2.5 experiments match the linear theory: label noise & spectral decay decide the winner.

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